ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

# MATHEMATICS

## PAPER 1 · PRACTICE D · MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central Grade 8 Mathematics paper 1 · practice d.**

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

**How to mark:** each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct answers.

## QUESTION 1

**1** 1.1 B ✓ 1.2 D ✓ 1.3 A ✓ 1.4 C ✓ 1.5 A ✓ 1.6 D ✓ 1.7 B ✓ 1.8 C ✓ 1.9 D ✓ 1.10 A ✓ [10]

## QUESTION 2

**2.1.1** $40 \times \text{R}2{,}50 = \text{R}100$ ✓ (1)

**2.1.2** $40 \times \text{R}4 - \text{R}100 = \text{R}160 - \text{R}100 = \text{R}60$ ✓ (1)

**2.1.3** $\text{R}60 \div \text{R}100 \times 100\%$ ✓ $= 60\%$ ✓ (2)

**2.2** Year 1: $\text{R}5\,000 \times 1{,}08 = \text{R}5\,400$ ✓; year 2: $\text{R}5\,400 \times 1{,}08$ ✓ $= \text{R}5\,832$ ✓ (3)

**2.3** Petrol $= 450 \div 100 \times 7 = 31{,}5\ \ell$ ✓; cost $= 31{,}5 \times \text{R}23{,}50$ ✓ $= \text{R}740{,}25$ ✓ (3)

**2.4** $\text{R}1\,000 \div \text{R}18{,}20$ ✓ $= \text{US\$}54{,}95$ ✓ (2)

**[12]**

## QUESTION 3

**3.1** (a) $4a^6 \times 3$ ✓ $= 12a^6$ ✓ (b) $x^8 \div x^4$ ✓ $= x^4$ ✓ (c) 4 ✓ $x^4$ ✓ (6)

**3.2** (a) $3{,}45 \times 10^6$ ✓ (b) $6{,}2 \times 10^{-5}$ ✓ (2)

**3.3** $-18 + 7 - 12$ ✓ $= -23$, so 23 m below sea level ✓ (2)

**[10]**

## QUESTION 4

**4.1** (a) $^7/_3 - {}^7/_4 = {}^{28}/_{12} - {}^{21}/_{12}$ ✓ $= {}^7/_{12}$ ✓ (b) $^3/_5 \times {}^{10}/_9$ ✓ $= {}^2/_3$ ✓ (c) 0,06 ✓ (5)

**4.2** Walk: $^3/_8 \times 120 = 45$ ✓; taxi: $0{,}25 \times 120 = 30$ ✓; bus: $120 - 45 - 30 = 45$ learners ✓ (3)

**[8]**

## QUESTION 5

**5.1.1** $y = 3x$ ✓ $+ 2$ ✓ (2)

**5.1.2** $a = 3(10) + 2 = 32$ ✓ (1)

**5.1.3** $3b + 2 = 62$, so $b = 20$ ✓ (1)

**5.2** (a) $4x - 3x + 15$ ✓ $= x + 15$ ✓ (b) $2a$ ✓ $- 3b$ ✓ (4)

**5.3** $(-2)^2 - 2(-2)(3) + 3^3$ ✓ $= 4 + 12 + 27$ ✓ $= 43$ ✓ (3)

**5.4** (a) $3x = 15$ ✓; $x = 5$ ✓ (b) $3x - 12 = -6$ ✓; $x = 2$ ✓ (c) $^{x}/_{4} = 3$ ✓; $x =$ 12 ✓ (6)

**5.5** Width $= w$, length $= w + 5$: $2(w + w + 5) = 46$ ✓; $4w + 10 = 46$, $w = 9$ ✓; width 9 cm and length 14 cm ✓ (3)

**[20]**

## QUESTION 6

**6.1** (a) $2x + 10° = 3x - 20°$ ✓ (vertically opposite angles) ✓; $x = 30°$ ✓ (4)
(b) $\angle AOC = 70°$, so $\angle AOD = 180° - 70° = 110°$ (angles on a straight line) ✓

**6.2** (a) $\angle PQN = 65°$ ✓ (alternate angles, KL ∥ MN) ✓ (4)
(b) $\angle PQM = 180° - 65° = 115°$ ✓ (co-interior angles, KL ∥ MN) ✓

**6.3** EF ÷ BC = DE ÷ AB = 10 ÷ 4 ✓, so EF = 6 × 2,5 = 15 cm ✓ (2)

**6.4** (a) Square ✓ (b) Trapezium ✓ (c) Rhombus ✓ (3)

**6.5** $x + 2x + 150° = 360°$ ✓ (angles around a point) ✓; $3x = 210°$, $x = 70°$ ✓ (3)

**[16]**

## QUESTION 7

**7.1** $h^2 = 5^2 - 1{,}4^2$ ✓ $= 25 - 1{,}96 = 23{,}04$ ✓; $h = 4{,}8$ m ✓ (3)

**7.2** Area of the triangle $= {}^{1}/_{2} \times 6 \times 4 = 12$ cm² ✓; volume $= 12 \times 10$ ✓ $= 120$ cm³ ✓ (3)

**7.3** (a) C $= \pi \times d = {}^{22}/_{7} \times 14$ ✓ $= 44$ cm ✓ (b) A $= \pi r^2 = {}^{22}/_{7} \times 7 \times 7$ ✓ $=$ 154 cm² ✓ (4)

**7.4** (a) 2,5 × 10 000 ✓ = 25 000 cm² ✓ (b) 3 500 ÷ 1 000 ✓ = 3,5 ℓ ✓ (4)

**[14]**

## QUESTION 8

**8.1.1** 74 ✓ (1)

**8.1.2** (71 + 74) ÷ 2 ✓ = 72,5 ✓ (2)

**8.1.3** 98 − 45 = 53 ✓ (1)

**8.1.4** 840 ÷ 12 ✓ = 70 ✓ (2)

**[6]**

## QUESTION 9

**9.1.1** $A'(-2\,;0)$ ✓ $B'(1\,;0)$ ✓ $C'(-2\,;3)$ ✓ (3)

**9.1.2** Yes – a translation does not change the size or shape of a figure ✓ (1)

**[4]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**